

Beginning CI/CD Pipeline using GitHub!

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About Me



Lead DevOps
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LFCE, LFCS
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Open Source
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Agenda

! What are CI/CD
Pipelines?



Why do we need



it? Github Actions



Demo

Q&A

What are CI/ CD Pipelines?

Full Form of CI/CD

Continuous **I**ntegration

Continuous **D**elivery (or **D**eployment or both)

CI/CD Pipelines

A CI/CD pipeline is a **series of**  that must be **performed** in order to deliver a **new version**  of software.

A CI/CD pipeline introduces **automation**  to improve the process of application **development**, **integration**  and **testing**  phases, as well as **at the** **live**  and **deployment**  during **deployment**  .

Elements of a CI/CD pipeline



Build - Compile the code



Test - Test features, measure coverage



Release - Deliver artifacts to repository



Deploy - Load the artifacts into production environment

Elements of a CI/CD pipeline



Build



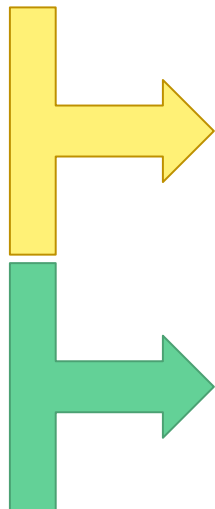
Test



Release



Deploy



Continuous Integration

Continuous
Delivery/Deployment

**Why CI/CD
Pipelines are
important?**

Importance of CI/CD pipelines



Reduce manual works for Ops team



More test coverage and fault

detection Faster release and happy customer

Github Actions

Git is a popular SCM
(Source Code
Management) tool

Github is a popular
hosting service for Git
and more

Github Actions are
powerful automation tools
provided by Github to
meet all your CI/CD needs



GitHub Actions

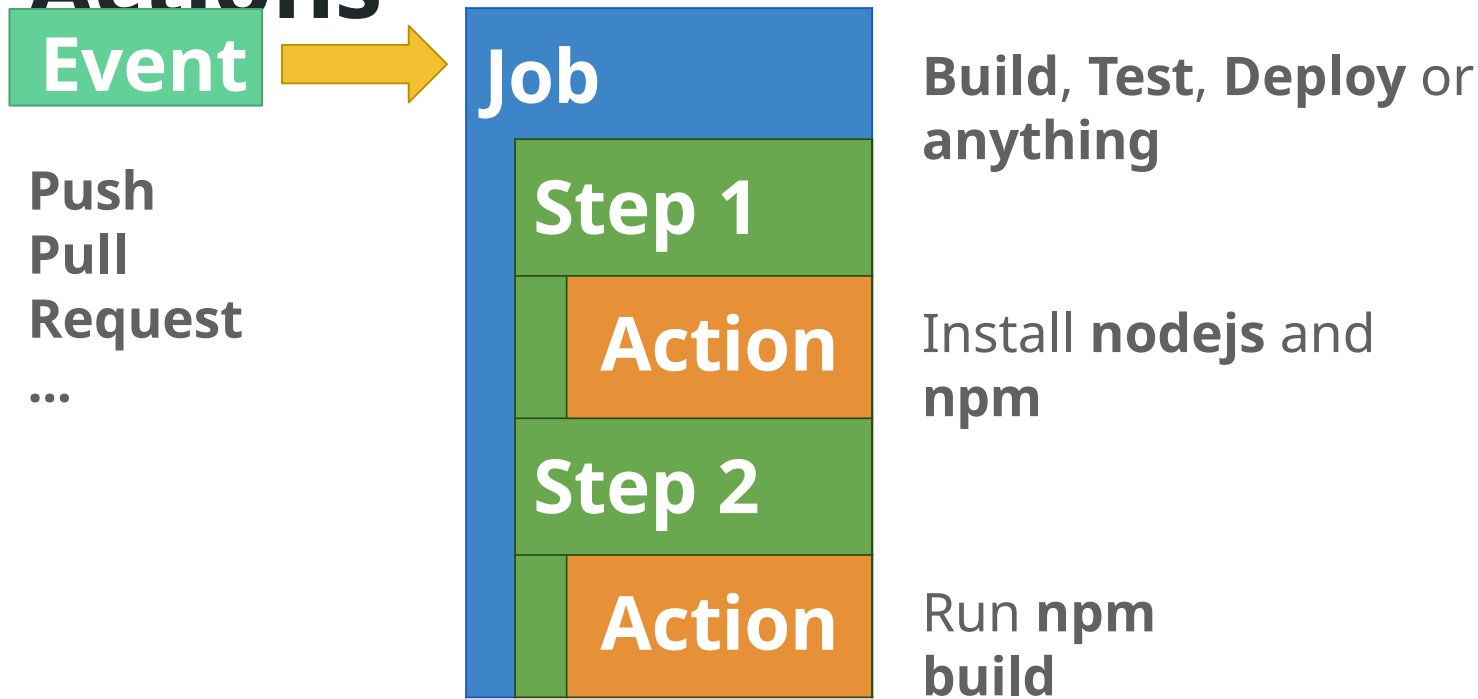


Github Actions

- Introduced in late 2018
- YAML based
- Community-driven
- Runs Win, Linux, Mac
- Attractive Pricing

Components of Github

Actions



Demo

Workflow File

```
name: My CI
Workflow on:
  push
:
jobs:
  build:
    runs-on: ubuntu-
    20.04 steps:
      - name: Clone repository
        uses:
          actions/checkout@v2
      - name: Set up Node.js
        uses: actions/setup-
          node@v2
      - name: Install npm
        dependencies uses: npm
        install
      - name: Build
        artifacts run: npm
        run build
      - name: Archive
        artifacts
```

Workflow Events

```
name: My CI
Workflow on:
  push:
    branches:
      - master
      - release
    /* tags:
      - v1.*
  pull_request
    : branches:
      - mai

n
scheduled: '0
0 * * * '
workflow_dispatch:
...
```


Workflow Jobs

```
name: My CI
Workflow
jobs:
  build:
    runs-on: ubuntu-
    20.04 steps:
      - name: Clone repository
        uses:
          actions/checkout@v2
      - name: Set up Node.js
        uses: actions/setup-
          node@v2
      - name: Install and build
        run: npm install && npm run
  build deliver:
    runs-on:
      ubuntu-20.04 steps:
      - name: Push to
        remote run: some-
          command
```

Workflow Steps and Actions

```
name: My CI
Workflow
jobs:
  build-linux:
    runs-on: ubuntu-
    20.04 steps:
      - name: Set up
        Node.js
        uses:
        actions/setup-
        node@v2
      - name: Run
        commands run:
        npm run build
  build-windows:
    runs-on: windows-
    latest steps:
      - name: Set up .NET
        uses:
        actions/setup-
        dotnet@v1
```

Dem

o

<https://github.com/beaccloudpro/github-cicd-demo>

**Q&
A**

